

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 26

Flipped Class: Clinical Cases and Imaging of the Head and Neck

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Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0064	Describe the fascial layers of the neck including investing fascia, pretracheal and buccopharyngeal fascia, prevertebral fascia and alar fascia and their clinical relevance
SOM.MKII.BPM2.1.NB.1.ANAT.0237	Describe the retropharyngeal space and its extension to the posterior mediastinum
SOM.MKII.BPM2.1.NB.1.ANAT.0235	Describe the third (danger) space and its extension to the through the posterior mediastinum to the diaphragm
SOM.MKII.BPM2.1.NB.1.ANAT.0239	Describe the anatomical relationships that are important for successful insertion of a catheter for central venous access and the possible complications of central venous access
SOM.MKII.BPM2.1.NB.1.ANAT.0240	Describe fractures of the orbit and their clinical consequences including extraocular muscle impingement
SOM.MKII.BPM2.1.NB.1.ANAT.0241	Describe the types of le forte facial fractures and the clinical consequences
SOM.MKII.BPM2.1.NB.1.ANAT.0243	Describe the nerves responsible for mediation of pain sensation during headaches
SOM.MKII.BPM2.1.NB.1.ANAT.0246	Describe the anatomical location, mechanism and clinical sequences of tentorial herniation
SOM.MKII.BPM2.1.NB.1.ANAT.0245	Describe the causes and locations of intracranial hematomas including epidural hematoma, subdural hematoma and subarachnoid hemorrhage
SOM.MKII.BPM2.1.NB.1.ANAT.0190	Explain the consequence of extracranial lesions of the vagus nerve

Recommended readings

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 4th ed. Elsevier. Chapter 8.

Siegel and Sapru Essential Neuroscience 4th ed. Chapter 13.

Neuroanatomy in clinical context. An atlas of structures, sections, systems and syndromes. Haines DE. 10th ed. Wolters Kluwer. Chapter 3



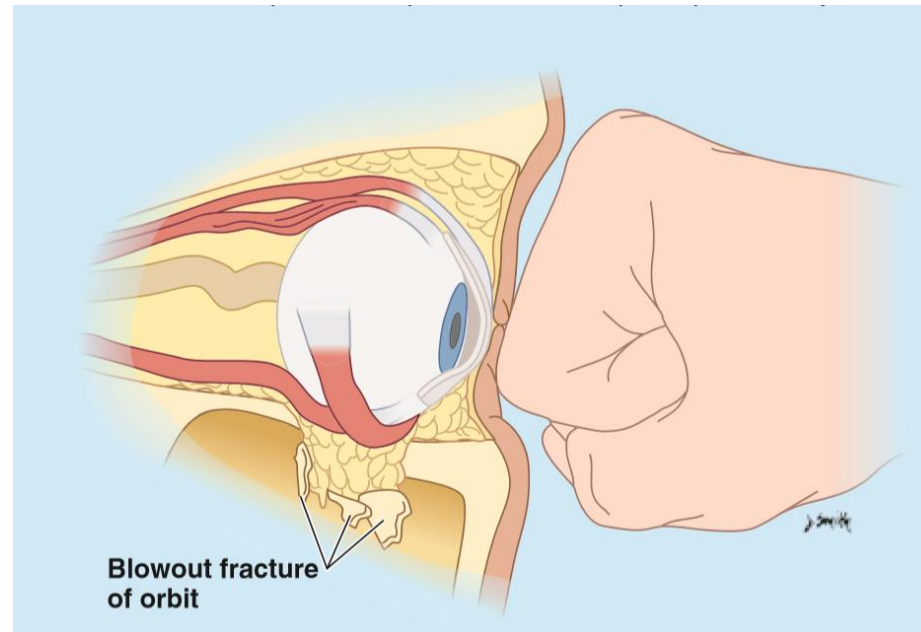
Pay special attention to the green 'In the Clinic' boxes which contain the clinical correlates for this topic.

Case 7

A 23- year-old man is brought to the emergency department because of severe left eye pain due to a night club fight 1 hour ago. His temperature 101.0°F, heart rate is 65/minute, respiration 20 breaths/minute, blood pressure 110/70 mmHg.

Physical examination shows left eye displaced posteriorly into sockets, with limited eye movement and loss of sensation along the trigeminal (V2) nerve distribution.

Which nerves fibers and muscles can be involved?

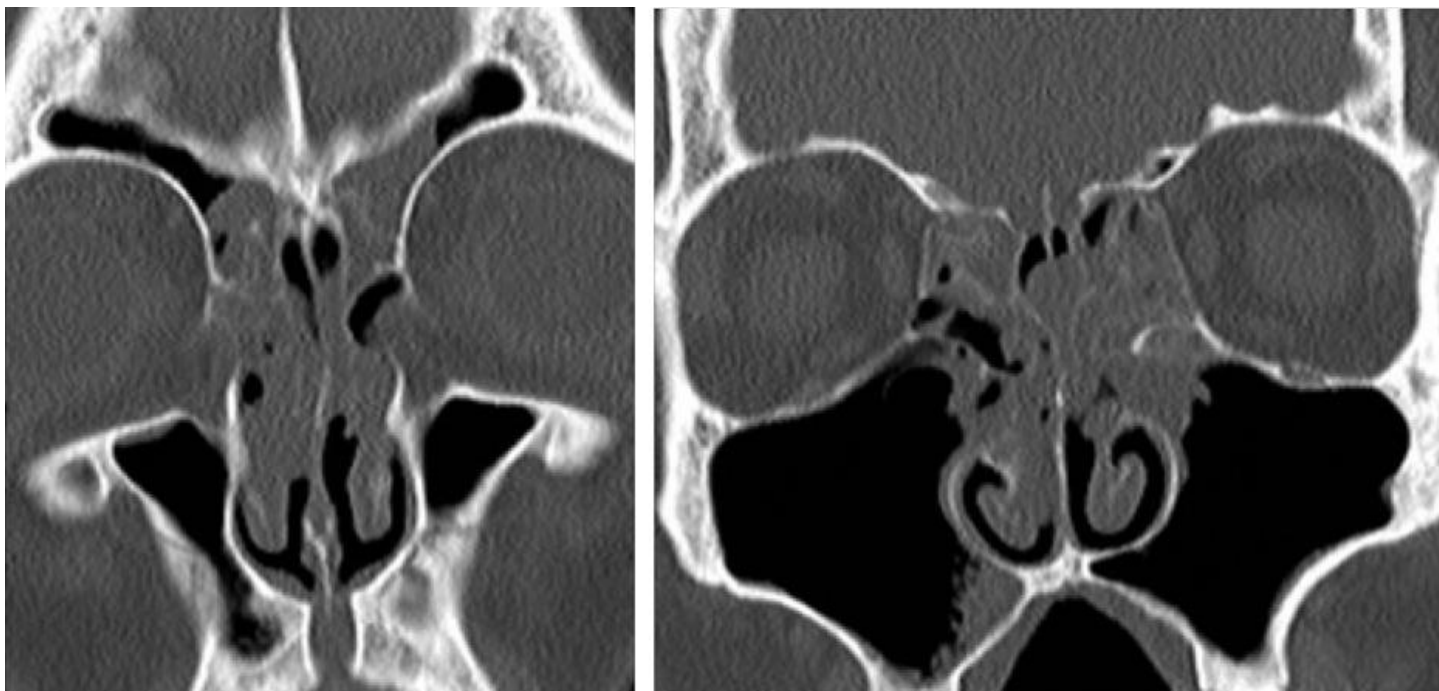


Case 8

A 28-year-old man comes to the physician because of frontal headache, sore throat and nasal congestion.

He is diagnosed with the common cold and is given paracetamol and a decongestant nasal spray. Two months later he returns with frontal headache, impaired nasal breathing and reduced sense of smell

Computed tomography (CT) of the paranasal sinuses showed nasal polyps confined to the ethmoid sinuses



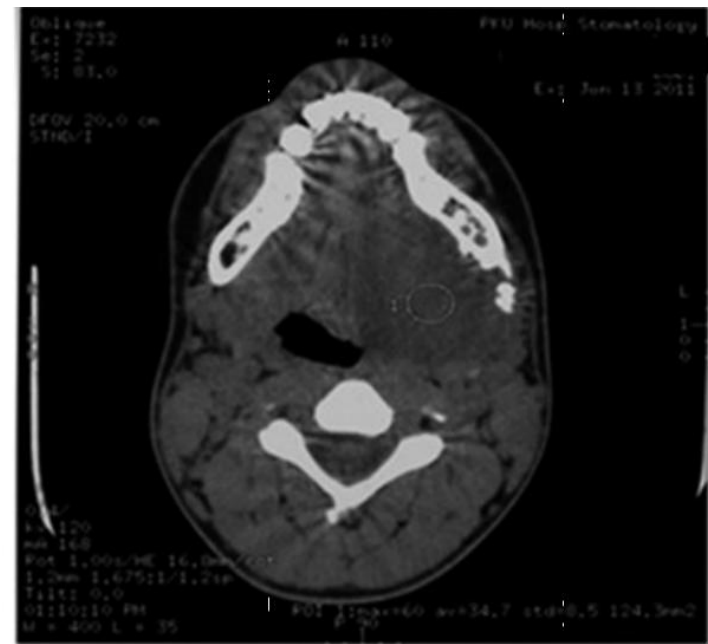
Case 9

A 51-year-old man is brought to the hospital for extraction of mandibular left third molar. Seven days after the surgery, he developed high grade fever, facial edema, intraoral purulent discharge and extreme local pain.

Physical examinations shows redness and swelling in the left side of the neck, which is tender and hot on palpation.

Lab results: hematocrit (36.2%), hemoglobin (11.9 g/dL); leukocytes (24500/mm³), C-reactive protein (18.15 mg/L) and erythrocyte sedimentation rate (20 mm/h).

How does the infection spread?





Case 10

- A 43-year-old Female comes to the physician because of shortness of breath, cough, and dysphagia.
- *Her vitals are: temperature 37.2°C (99 °F), pulse 98/min, blood pressure 110/75 mm Hg, respiration 20/min.*
- *In a physical examination, the palpable thyroid gland was found to be diffusely swollen and soft and exhibited poor mobility. The lower pole of the thyroid was not palpable.*
- *A blood examination revealed normal thyroid function. Ultrasound showed that the cervical thyroid gland was diffusely enlarged and exhibited multiple regions of cystic degeneration, but no obvious malignant findings were observed.*

Case 11

An 18-year-old man comes to the physical because of 1-hour history of trauma to the right temporal region. He briefly loses consciousness. Physical examination shows GCS (Glasgow Coma Scale) of 15. 4-hours following admission, GCS deteriorates to 8 and his left pupil is dilated.

1. What is the most likely differential diagnosis?
2. What is the anatomical basis for the left pupillary dilatation?
3. Where would you locate the middle meningeal artery for the purpose of making a burr hole?
4. What layers of the scalp would you encounter in your incision?

